



COURSE SYLLABUS						
Course Code	COMP 014	Course Title	Quantitative Methods With Modeling And Simulation		Course Credit	3
Course Description						
Pre-Requisites			Co-Requisites	None		
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				
INSTITUTIONAL LEARNING OUTCOMES (ILO)		As a polytechnic state university, PUP shall develop its students to possess: 1. Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. 2. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. 3. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors. 4. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state-of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. 5. Passion for Lifelong Learning. Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.				



	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10. Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11. Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
		Apply knowledge of computing, science, and mathematics appropriate to the discipline.								
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution									
	Identify and analyze user needs and take them into account in the selection, creation, evaluation									



	and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									



	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)	Understand principles of simulation systems	Alignment to PLOs							
		1	2	3	4	5	6	7	8
	Develop conceptual models of complex systems								
	Create computer simulation models								
	Analyze and evaluate simulation results								





	Implement IT-based solutions effectively								
	Demonstrate problem-solving skills in optimizing processes using quantitative and simulation methods.								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	- Understand the fundamental principles of modeling and simulation. - Discuss the importance and applications of modeling in various fields.	Introduction to Modeling and Simulation	Lecture discussion; overview of course framework; introduction activity.	Assigned readings and online reflection.	Formative quiz; participation					
3	- Analyze system components and interactions. - Classify systems	System Analysis and Classification	System mapping activity.	Online quiz and group analysis.	Worksheet on system classification					



	based on structure and behavior.								
4	• Differentiate model classifications and their purposes. • Identify conceptual, abstract, simulation, and heterogeneous models.	Model Classification: Conceptual, Abstract, Simulation, Heterogeneous	Lecture and example-based discussions.	Interactive simulation examples online.	Short quiz; model identification task				
5	• Describe the role of simulation systems and languages in model execution. • Compare features of different simulation languages.	Simulation Systems and Languages	Demo of simulation platforms.	Watch recorded demo; short reflective essay.	Reflection paper; recitation				
6	• Construct simple parallel process models using Petri Nets and Finite Automata.	Parallel Process Modeling: Petri Nets, Finite Automata	Hands-on modeling exercise.	Online simulation tool practice.	Quiz #1; performance task				
7	• Explain and model queuing systems and continuous simulations.	Queuing Systems & Continuous Simulation Models	Queuing model demo and problem-solving.	Practice activities via LMS.	Worksheet submission				



8	Identify and apply different system models and special model classes.	System Models & Special Model Classes	Guided discussion with examples.	Case-based learning activity.	Midterm review quiz					
9	Conduct verification, validation, and analysis of simulation models.	Verification, Validation, and Analysis of Simulation Results	Group validation activity.	Online critique of sample simulations.	Midterm Exam					
MIDTERM WRITTEN EXAMINATION										
10	Combine multiple simulation techniques for problem-solving.	Combined Simulation Techniques	Workshop on hybrid modeling.	Reading assignment and reflection.	Group activity output					
11	Evaluate the role of simulation in digital systems design and testing.	Role of Simulation in Digital Systems Design	Lecture and case demonstration.	Online discussion of digital simulation tools.	Reflection paper					
12	Apply model optimization and experimental design in	Model Optimization & Experiment Design	Hands-on lab exercise on optimization.	Data analysis assignment.	Quiz #2; optimization report					



	simulation studies.									
13	• Demonstrate understanding of stochastic models and Monte Carlo methods.	Monte Carlo Methods & Stochastic Models	Simulation run demonstration.	Video lesson and practice quiz.	Simulation output submission					
14	• Visualize and interpret simulation results using interactive tools.	Simulation Results Visualization & Interactive Simulation	Visualization lab using Excel or Simulate.	Create visual dashboards online.	Visualization project					
15	• Examine the integration of virtual reality in simulation environments.	Virtual Reality & Simulation	Demonstration of VR-based simulations.	Reading and reflection on VR applications.	Written reflection					
16	• Explain pseudorandom number generation and testing in simulation.	Pseudorandom Number Generation & Testing	Lecture and computation activity.	Online quiz and exercises.	Quiz #3					
17	• Compare and evaluate commonly used simulation systems and tools.	Overview of Commonly Used Simulation Systems	Hands-on demo; tool comparison.	Self-paced review of simulation platforms.	Evaluation matrix; participation					
18	FINAL WRITTEN EXAMINATION									



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92nd in Southeast Asia
541-560 in Asia



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RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI
APPLICATION

Rank 39
STUDENT SUPPORT
ENGAGEMENT

Rank 50
SYMBOL/
PROMOTION



STARS



Employability
Online Learning
Internationalization



Teaching
Academic
Development
Inclusiveness



Social Responsibility
Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)



Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+



	• Design and implement a simple modeling and simulation project.	Modeling and Simulation Application Project	Project presentation and defense.	Online submission of documentation.	Final Project Presentation					
18	FINAL EXAMINATION									
LIST OF REFERENCES BOOKS: WEBSITE: https://www.scribd.com/presentation/508926254/QUANTITATIVE-METHODS1 https://www.coursera.org/learn/wharton-quantitative-modeling										



CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.

Times Higher Education Impact Rankings 2024
DEBUTS AT 1501+



The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a





	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.	
COURSE REQUIREMENT/S	GRADING SYSTEM	
- Making a Learning Plan in Teaching Prose (focusing on Completeness, Appropriate Approach Used, and Facilitative Process) - Teaching Demonstration in Prose or poetry (focusing on Teacher-Like Simulation, Preparation, Classroom Management, Learning Environment, and Interest and Diligence in Teaching Poetry)	Mid Term	Final Term
	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper	Class Standing 70% Exercises/Activities Summative Quizzes Recitation and Attendance Reflection Paper
	Midterm Examinations 30% 100%	Final Examinations 30% 100%
	Final Grade = (Midterm Grade + Final Term Grade) / 2	
CLASS INFORMATION		FACULTY INFORMATION
Course, Year and Section: BSIT 2-1		Name of Faculty: RONALD JOY TENGCO
Day and Time:		Consultation Time:
Room:		Office Tel. No./ Mobile Phone No.
School Year: 2	Semester: 1	E-Mail Address:



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APPLICATION

Rank 39
STUDENT SUPPORT
ENGAGEMENT

Rank 50
SYMBOL/
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STARS

★★★★★
Employability

★★★★★
Online Learning

★
Internationalization

★★★★★
Teaching

★★★★★
Academic
Development

★★★★★
Inclusiveness

★★★★★
Social Responsibility

★★★★★
Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)



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